Lecture Transcript by Slide

Week 1 · Chapter 1: Electrophysiology · 27 min

Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

0:00:00 Chapter 1: Electrophysiology (title slide)

0:00:22 EKGs Are Graphic Representations of Electrical Activity

0:01:33 EKGs Reflect Electrical, Not Mechanical, Activity

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0:16:04 Inherent Rates

0:19:29 Electrical Conduction System (revisited)

0:20:47 Pacemaker Rule

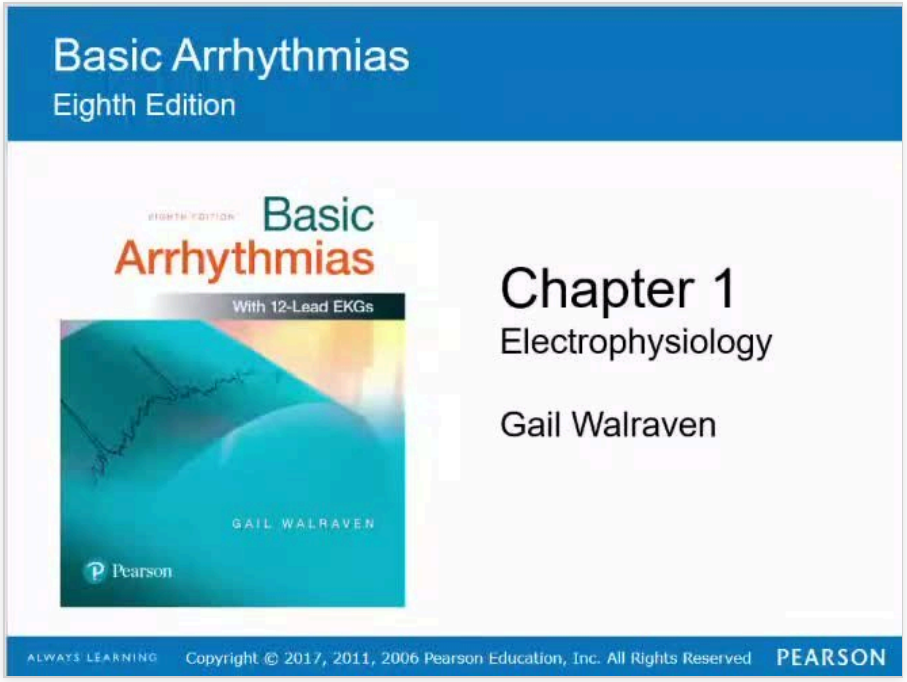
0:22:50 Irritability

0:24:14 Escape Mechanism

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1 Chapter 1: Electrophysiology (title slide)

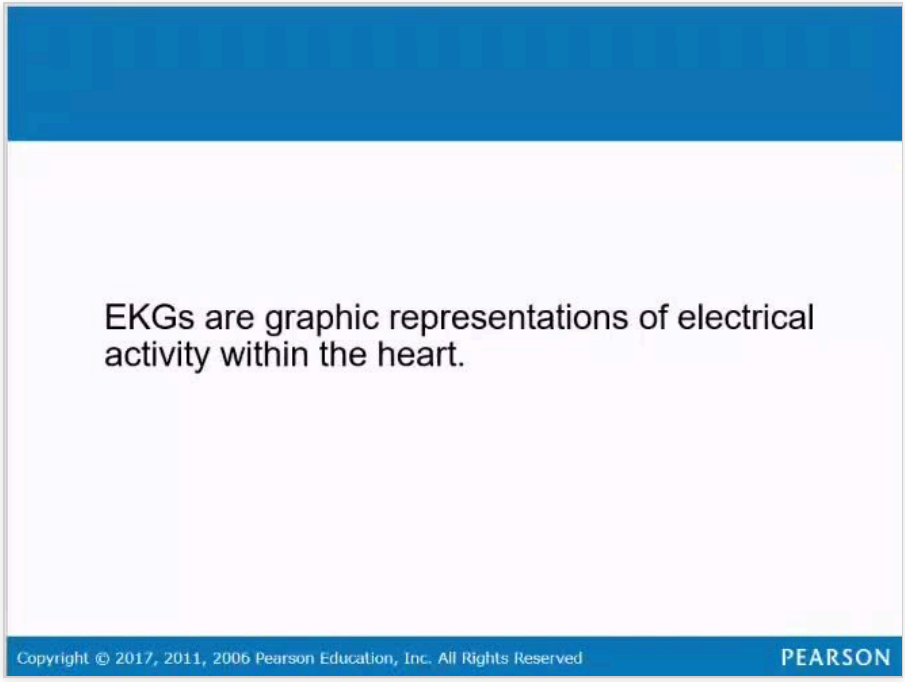
0:00:00 – 0:00:22



0:00:00 Okay, in our second lecture for REST 101 Spring 2024 semester, we're going to talk about electrophysiology, cardiac electrophysiology. So we'll talk briefly here about EKGs and what they are.

2 EKGs Are Graphic Representations of Electrical Activity

0:00:22 – 0:01:33

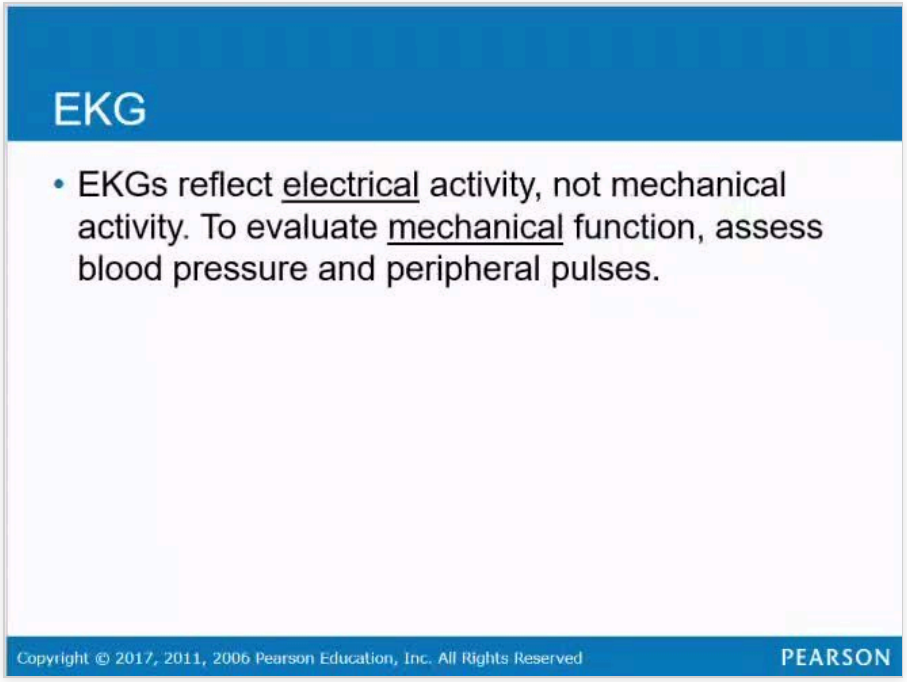


0:00:27 EKGs are electrical activity of the heart, and the impulses and subsequent contractions all create electrical activity that are recorded on an EKG, a 12-lead EKG, or recorded on a monitoring device. So EKGs are graphic representation of electrical activity within the heart.

0:01:08 So when we are looking at the electrical activity, if you're recording it on paper, we're looking and recording the electrical activity over time. It generally moves from the left side of the screen or the left side of the paper to the right side. So we're recording electrical activity over time. EKGs reflect only the electrical activity,

3 EKGs Reflect Electrical, Not Mechanical, Activity

0:01:33 – 0:02:45

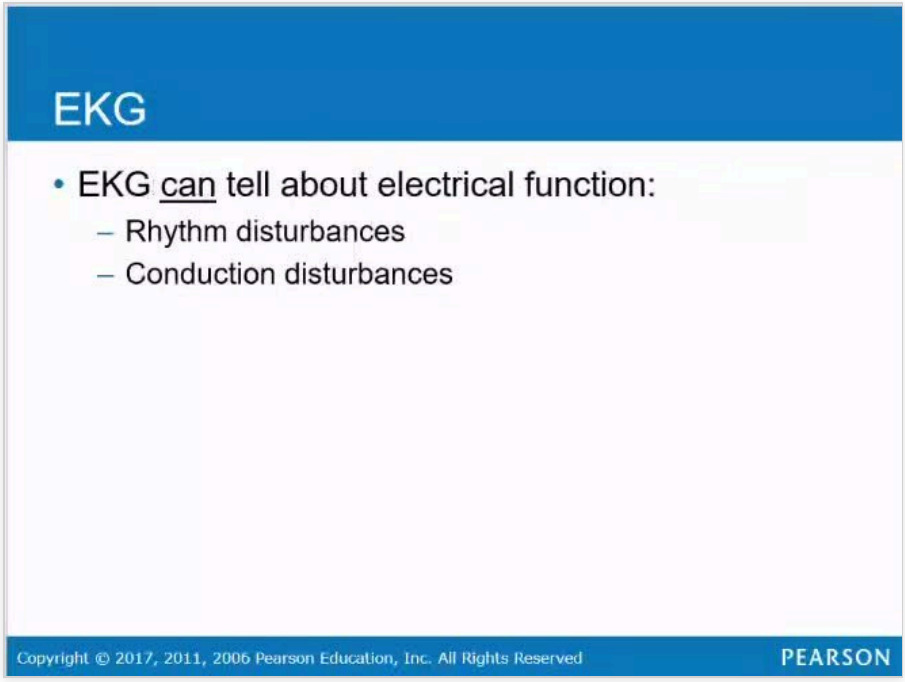


0:01:40 not the mechanical activity. The mechanical activity is referring to contraction of the atria or contraction of the ventricles. To assess the mechanical function, we would do bedside assessment. We would feel for peripheral pulses, a radial pulse or a brachial pulse.

0:02:12 And we would also measure their blood pressure. Because if there's no mechanical activity, there's no pumping of blood, they wouldn't have a pulse, and they wouldn't have a blood pressure. Or if the mechanical activity is not performing very well, the pulses may be weaker, and the blood pressure may be lower. So again, when we record this, we're only recording electrical activity. It does not assess any mechanical activity.

4 What an EKG Can Tell: Electrical Function

0:02:45 – 0:04:15

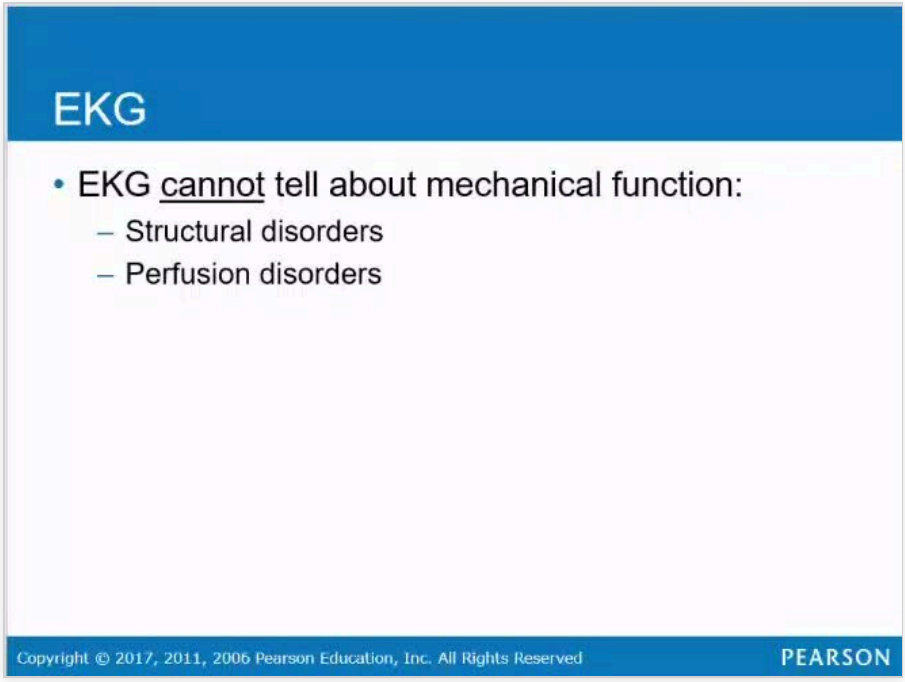


0:02:45 What EKGs can tell us about the electrical function is if there's a rhythm disturbance or a conduction disturbance. Rhythm disturbances can be an abnormal rhythm. An abnormal rhythm like atrial fibrillation or atrial flutter or a junctional rhythm. And the conduction disturbances can tell us if there's an abnormality somewhere along the cardiac pathway, the nerve pathway of the heart. If somebody was to have a heart attack, like I was talking about in the previous lecture, and some of the cells were affected because they didn't get enough oxygen for an extended period of time, we may see a disturbance in the conduction system, the heart's conduction system.

0:04:02 There may be a delay. There may be a change in the appearance of some of the waveforms. And so that's what I mean by a conduction disturbance. And so that's what I mean by a conduction disturbance. EKGs also do not tell us about mechanical function, as I mentioned.

5 What an EKG Cannot Tell: Mechanical Function

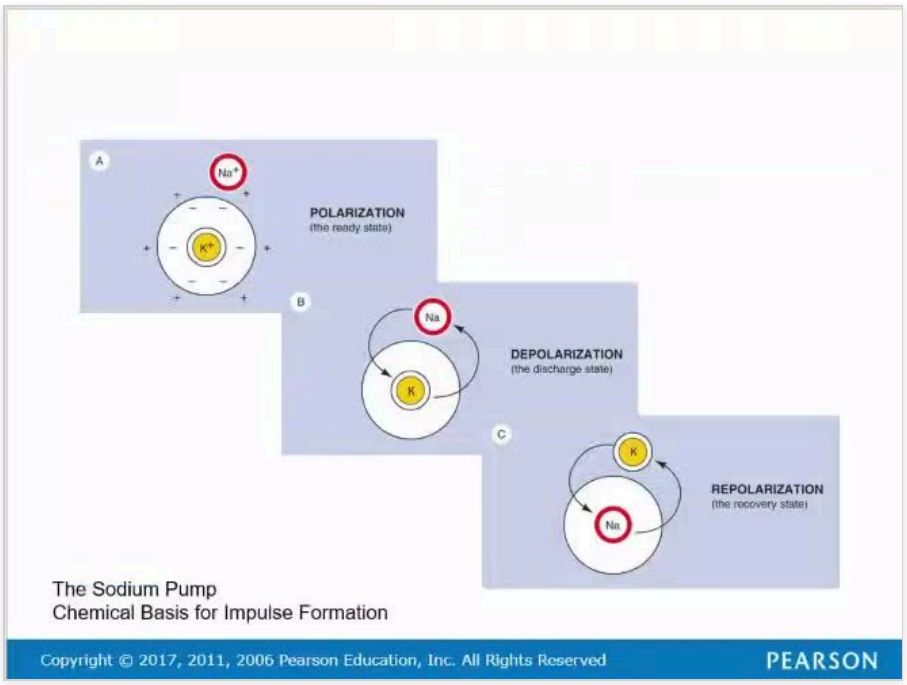
0:04:15 – 0:05:17



0:04:25 So mechanical function, whether the heart is pumping or not. But it also cannot tell us if there's a structural disorder, meaning an abnormality in the structure of the heart. Sometimes they can have a hole in the septal wall between the atria, known as an ASD, or a hole in the septal wall between the ventricles, known as a VSD. There's not supposed to be a hole there, but that would be an abnormality in the structure of the heart itself. EKGs also don't tell us about perfusion disorders. So if the blood is going in the wrong direction that it's not supposed to, that could be a perfusion disorder. We'll talk a little bit about polarization, depolarization, and repolarization.

6 The Sodium Pump: Chemical Basis for Impulse Formation

0:05:17 – 0:08:59



0:05:24 I'm not going to get too far into this. But when they sell, and these cells could be a cardiac muscle cell, or these cells could be a nerve cell. It doesn't really matter. It could be a cell anywhere in your body. When a cell is reset, it's sitting in what we call a polarized or ready state. Now, normally in the ready state, the cell is slightly negatively charged inside and slightly positively charged outside the cell. And the majority of the potassium sits in the cell. The majority of the sodium sits outside the cell. When an impulse comes along, that impulse has to get the individual cell to the threshold.

0:06:25 Once we reach the threshold, the cell will then depolarize. During depolarization, the sodium channels in the cell wall will open up, and sodium rushes in. The potassium channels will also open, and the potassium slightly leaks out. So this is known as depolarization. When a stimulus meets the threshold of a cell, it begins to depolarize. Sodium rushes in, and potassium rushes out. And that goes from cell to cell, or it also goes to a number of cells that work at the same time in unison, like a chamber contraction. What causes depolarization is these electrolytes like sodium and potassium flow from a high concentration to a low concentration.

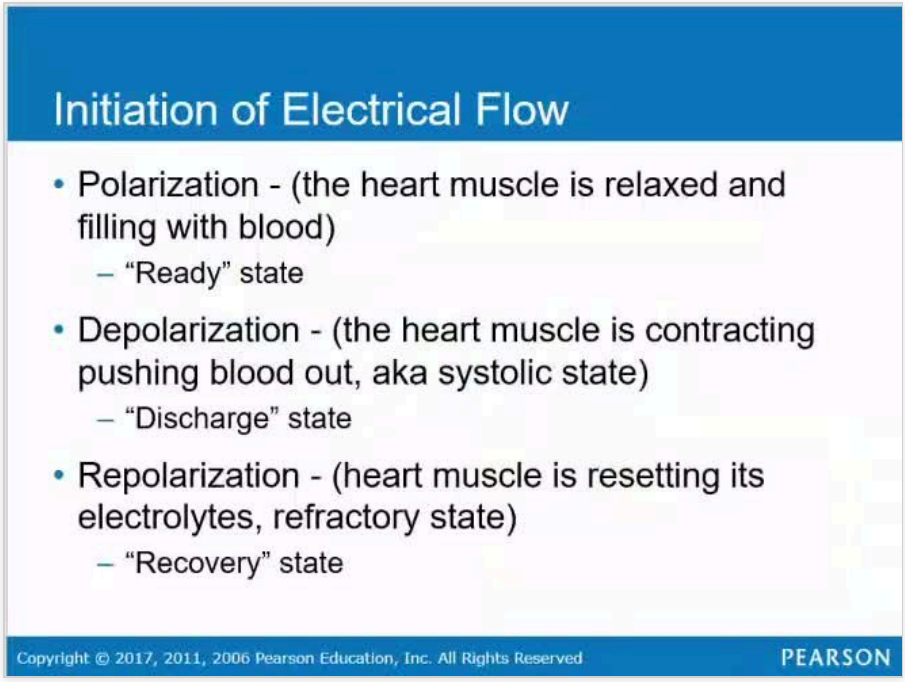
0:07:36 So there's not a lot of sodium inside the cell. The majority of it sits outside the cell. So when the threshold is reached, the sodium goes from outside the cell inside, high to low concentration, and the same with the potassium. A lot of potassium inside the cell, not a lot of potassium outside the cell. So then the potassium leaks out, and thus the cell depolarizes. And

this depolarization on the nerve cells continues to send the impulse along the pathway. This depolarization on a cardiac muscle cell causes contraction of the cell. Then the cell needs to reset itself. It does that through what's called the sodium-potassium ATPase pump.

0:08:26 When this happens, the cell is pumping the sodium back out of the cell, bringing the potassium back in the cell, trying to reset itself. This is known as the repolarization or the recovery state. Once the cell is completely repolarized, it goes back to the ready state, awaiting the next impulse to come along to meet the threshold and cause depolarization again. So we talked about these terms a little bit in the previous lecture.

7 Initiation of Electrical Flow

0:08:59 – 0:10:57

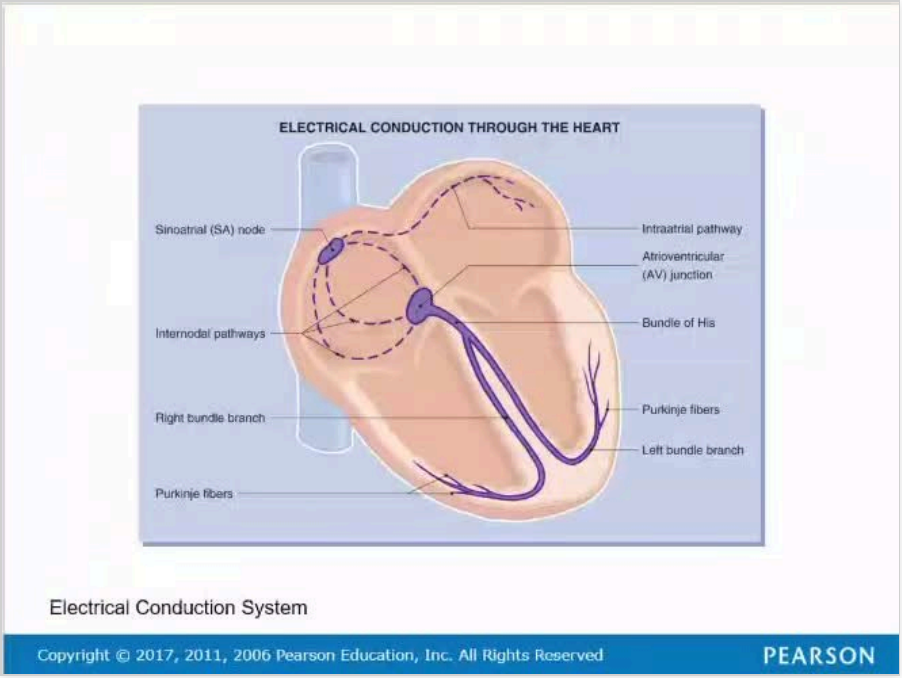


0:09:05 They're important terms to understand. Polarization means that the cells, or the majority of the cells, are sitting in what's called a ready state. So the heart is relaxed and filling with blood. So polarization, relaxation, ready state, or diastole all fit in that same category of terminology. So understand that when the heart is in the polarized state, it's relaxed, filling with blood, also known as diastole. During depolarization, we just discussed what happens during depolarization, the cells are contracting, pushing out blood.

0:10:05 So during contraction, this is also known as a systolic state. So depolarized, systolic, contraction, all in the same category of terminology. And then repolarization, where the heart is resetting the electrolytes. During that resetting of the electrolytes, there's something called the refractory state. So when the cells are depolarized, it's refractory or can't respond to another stimulus until it gets to a certain point. And we'll discuss the refractory state in a few slides. Okay, this is super important.

8 Electrical Conduction System

0:10:57 – 0:16:04



0:11:00 So this is the heart's electrical conduction system, the anatomy of the conduction system. And we'll talk a little bit about the function of the conduction system. Super important. There's a lot of information coming from this slide and subsequent slides after this that will help set the basis of understanding of rhythm interpretation throughout the rest of this course. So the heart has a specialized group of cells, their nerve cells, in this conduction system.

0:11:44 And the heart has a primary pacemaker. And that primary pacemaker is called the sinoatrial node or the SA node. So that is the primary pacemaker of the heart's conduction system.

0:12:04 And these are a group of cells that have their own automaticity. Automaticity is really referring to the fact that it can stimulate an impulse on its own. You don't have to think about stimulating an impulse for your heart to contract. These specialized group of cells do that without any input. That's what's referred to as automaticity. When the SA node fires and creates an impulse, it sends the electrical signal along two different pathways. It sends the impulse along what's called an intraatrial pathway. And this pathway delivers that impulse to the atrial muscle cells, both right and left atrial muscle cells.

0:12:59 Deliver that impulse to the atria for atrial contraction. At the same time the impulse is traveling on that pathway, it also travels on another pathway known as the internodal pathway. So the next node after the SA node is the AV node, AV standing for atrial ventricular node, also known as the AV junction or atrial ventricular junction. So it's the junction of electrical cells in between the atria and the ventricles. It typically sits right at the top of the bottom of the right

atrium, top of the right ventricle. Now the impulse gets sent to the AV node through that intranodal pathway. The AV node catches that impulse and it holds on to it for a second, for milliseconds actually.

0:14:03 And the reason why it holds on to it for milliseconds is because we want that electrical stimulus to be delivered to the atrial muscle cells, have the atria contract because we want the heart to be from the top down. That's when it is most effective. So we want the blood content in the atria to get pushed down to the ventricles prior to ventricle contraction. So once that occurs, now the AV node will release that impulse down to, this is still part of the junction, but this single branch below the AV node is called the bundle of his, or some people refer to as the his bundle.

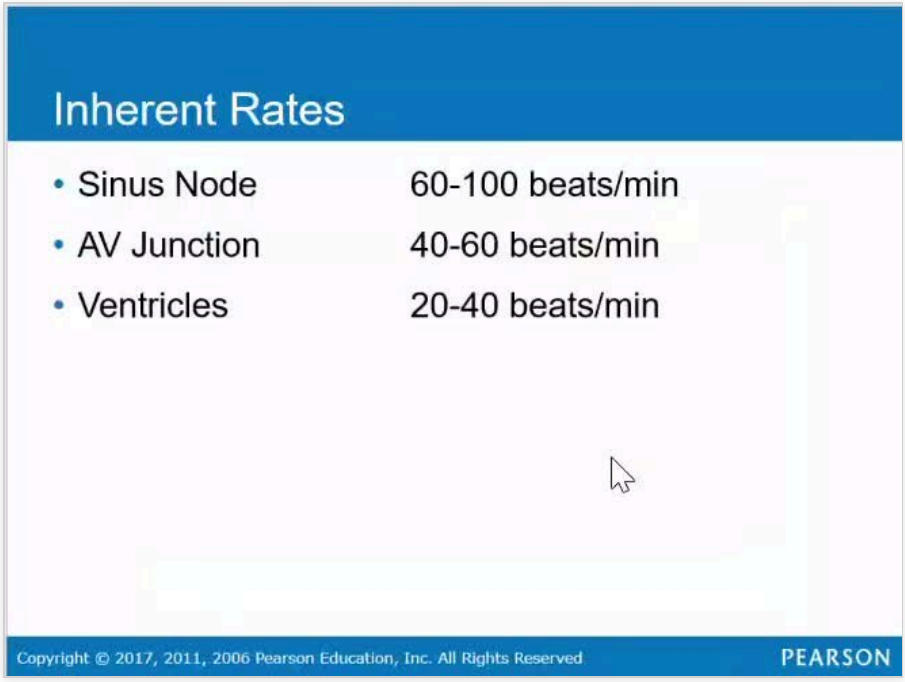
0:14:49 The bottom of that his bundle, it splits into what's called the right and left bundle branches. So this is a bundle of nerves. There's a left bundle branch and there is a right bundle branch. And these bundle branches continue to bring that electrical impulse out to these distal fibers known as the Purkinje fibers. These Purkinje fibers are the fibers that deliver the impulse to the ventricular muscle cells causing ventricular depolarization or ventricular contraction.

0:15:32 It works in unison. So it sends, the SA node sends a signal from one node to the next. The next node holds on to it. We have atrial contraction pushing their blood count down into the ventricles. Then the AV node lets it go. It travels through the rest of the electrical conduction system for ventricular contraction.

0:16:02 Now the SA node normally beats at 60 to 100 beats per minute.

9 Inherent Rates

0:16:04 – 0:19:29



0:16:12 So that's the number of impulses that the SA node normally generates in a healthy heart per minute. Heart rate is always in beats per minute.

0:16:25 The SA node actually can fire less than 60, heart rate less than 60. And that is a slow heart rate, below normal heart rate. We call that bradycardia. So a heart rate less than 60 is termed bradycardia. It can also fire above 100. And when the heart rate gets above 100, we refer to that rate as tachycardia. So we can have normal sinus rhythm, 60 to 100. We can have sinus bradycardia, a heart rate less than 60. And we can have sinus tachycardia, a heart rate above 100.

0:17:08 That's the primary pacemaker. Now, there are times when people have heart disease where the SA node doesn't function appropriately. It either slows down significantly, because once your heart rate gets down to a certain point, you don't have enough cardiac output to sustain normal daily activity. And remember, cardiac output is heart rate times stroke volume. Anytime you have a decrease in stroke volume, you'll have a decrease in cardiac output, you'll have a decrease in blood pressure. If you have a significant decrease in heart rate, which now lowers your cardiac output, you're going to have less cardiac output and your blood pressure is going to go down.

0:17:54 But if something happens to the SA node and it's not functioning, or it stops working entirely, we have what's called the backup pacemaker. The first backup pacemaker of the heart is called the AV junction, or the AV node. So if something happens to the SA node and its heart rate

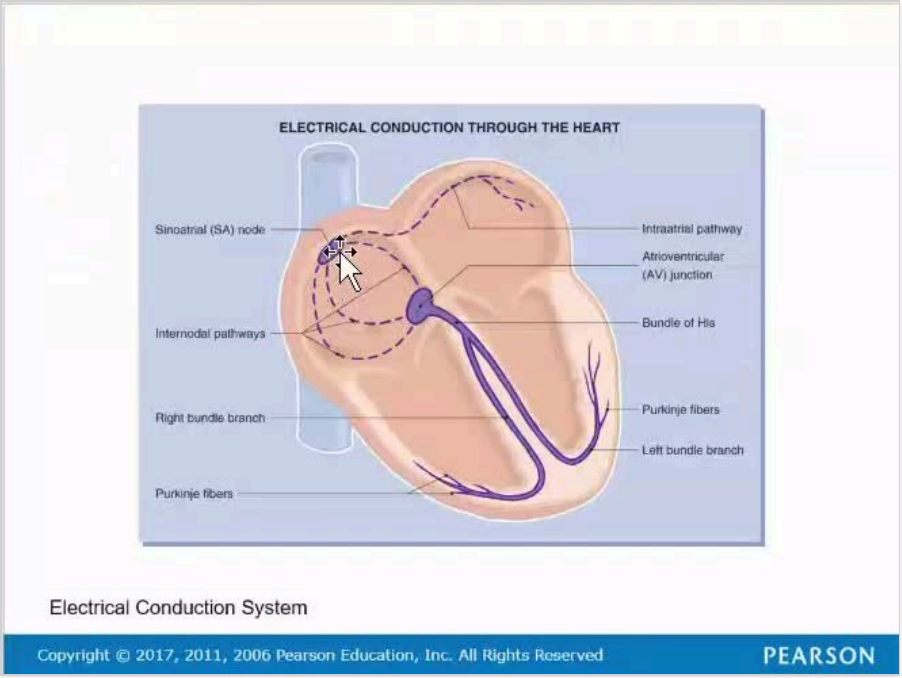
slows down below 40, the AV node will now take over for pacemaking responsibilities. And the inherent rate of the junction is 40 to 60 beats per minute.

0:18:27 Inherent rate, the junction can actually beat faster than 60 beats per minute. It can go 60 to 100, it can go above 100. So the AV junction itself can have different rates, but its inherent rate is 40 to 60. Now, if something were to happen to the AV node, in addition to the SA node, the last backup pacemaker that the heart electrical system has is somewhere along the conduction pathway, down in the ventricles. And the inherent rate of the ventricles is 20 to 40 beats per minute. Typically not fast enough to sustain normal activity. So it's usually a life-threatening rhythm because the heart rate is so slow, and the heart beats from the bottom.

0:19:20 It doesn't beat from the top down. So I'm going to go back a slide for a second here just to show you that SA node is the primary pacemaker.

10 Electrical Conduction System (revisited)

0:19:29 – 0:20:47

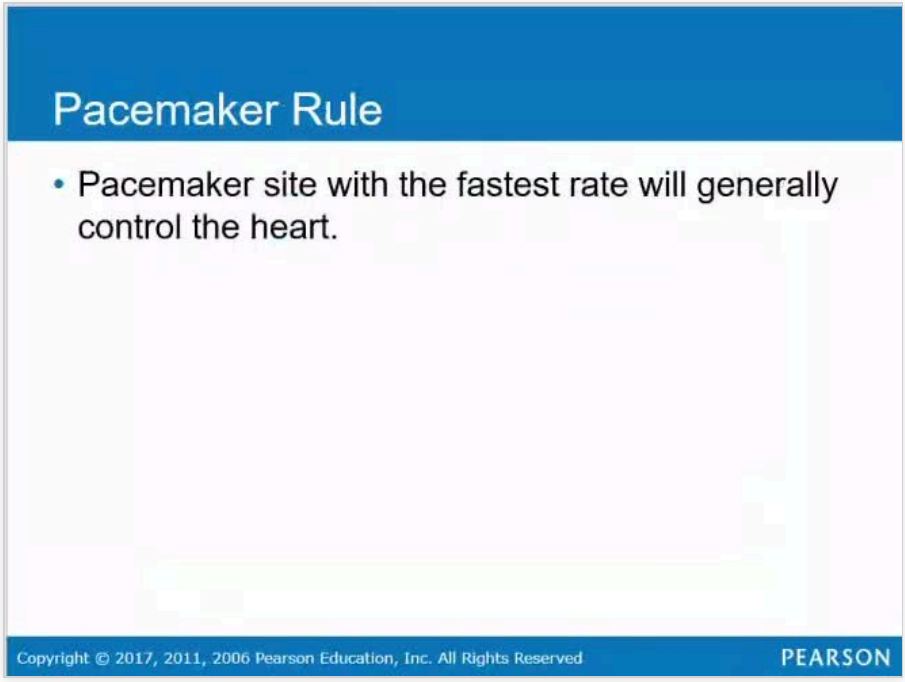


0:19:32 It's important that you know that. It's important that you know the pathway of the conduction. The AV node, I'm sorry, the SA node to the atrial pathway, intranodal pathway to the AV node, to the bundle of his, to the right and left bundle branches, to the Purkinje fibers. It's important that you know that the SA node normally fires 60 to 100 beats per minute. It's important that you understand the inherent rate of the AV node if it has to become the pacemaker in a situation where the SA node is not functioning. And it's also important that if the AV node fails, somewhere down here along the pathway, down in the ventricles, is one of these cardiac muscle cells will kick in as the last backup pacemaker, and it will fire inherently at 20 to 40 beats per minute.

0:20:34 Again, this is information that's going to set the basis or rules or guidelines for our ability to interpret rhythms as we go deeper into the semester.

11 Pacemaker Rule

0:20:47 – 0:22:50

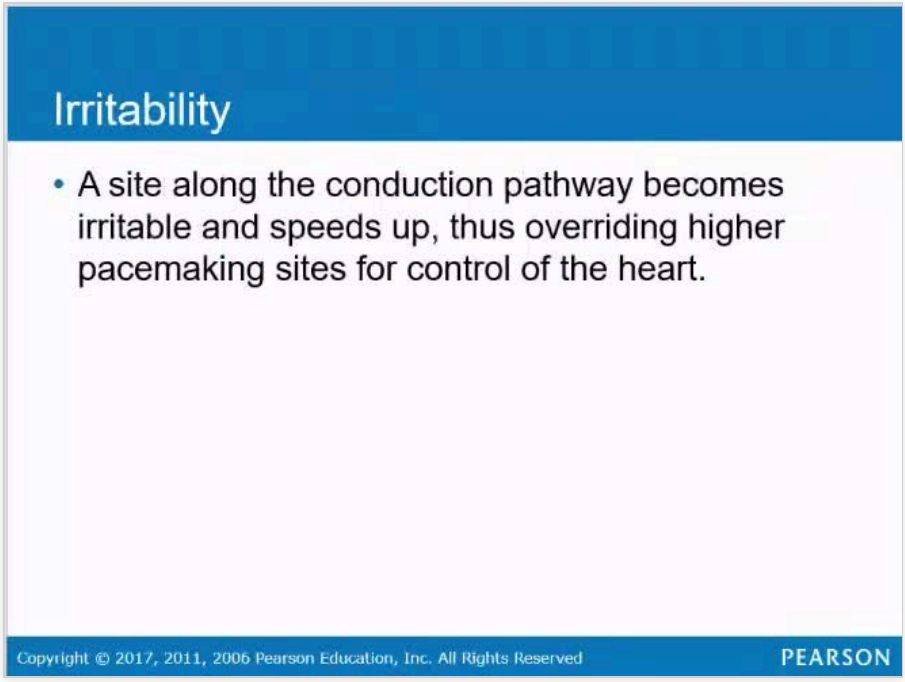


0:20:48 Okay, we're going to talk about pacemaker rule. So I just talked about three different pacemaking sites that the heart has. And I refer to them as the primary, the first backup, the last backup. And as we go from primary to backup, the rate inherently slows down. To the last backup, the rate inherently slows down even more. There are certain situations that can make these backup pacemakers go faster than their inherent rate. So if somebody has cardiac disease and part of the pacemaker is not getting enough oxygen or is sick or ill, it can cause those cells to beat or fire at a faster rate than its inherent rate.

0:21:47 So if the A-B node itself became irritable and started firing impulses, once the rate of those impulses is faster than the S-A node, it will now take over as the pacemaker. So this pacemaker rule is whatever pacemaker is firing at the fastest rate, that's the one that will control the heart. So when the heart's not irritable, those other pacemaker sites aren't going to fire. So the S-A node is going to be the pacemaker. But in a situation where the S-A node or somewhere down in the ventricles becomes irritable due to heart disease, and the irritability causes those sections of the heart to generate impulses faster than the S-A node, it's going to take over as pacemaker. So fastest rate wins. I was just talking about irritability.

12 Irritability

0:22:50 – 0:24:14

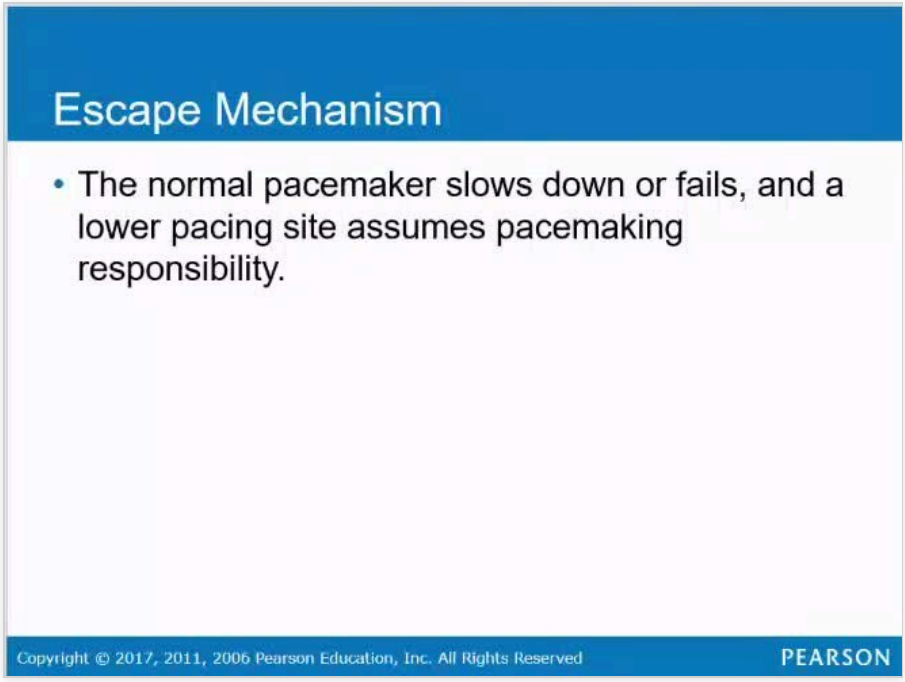


0:22:52 Irritability is a site along the conduction pathway that becomes irritable, either from disease or lack of oxygen. And once the cells there become irritable, it can start to generate impulses, and it can speed up the rate at which it fires these impulses. And if its rate is faster than the S-A node, it now becomes the pacemaker. So we could have a situation where the S-A node is functioning properly, but if somewhere in the A-V node becomes irritable, let's say the S-A node is firing at a normal heart rate of 80, and the S-A-V node becomes irritable, and it starts firing impulses at a rate of 20, 120, now the A-V node will take over even though the S-A node is firing appropriately.

0:23:44 And then we'll have a junctional rhythm that we're going to have to interpret. So irritability, any site along the pathway that becomes irritable and fires at a faster rate than the inherent pacemaker, that irritable spot is now going to take over for control of the heart. So it's important that you understand this concept. This is going to reappear over and over again as we start to interpret rhythms.

13 Escape Mechanism

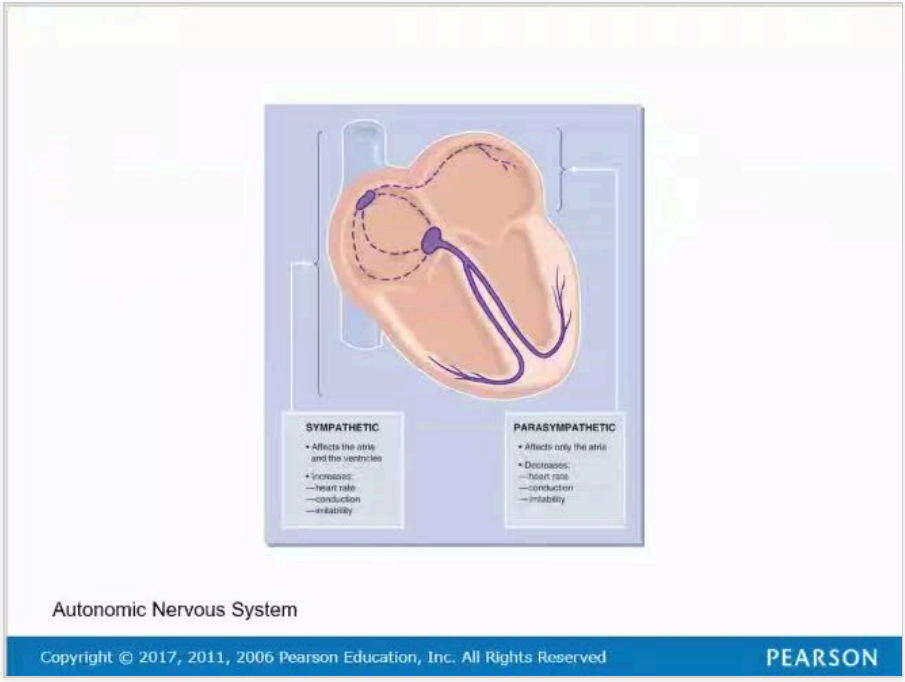
0:24:14 – 0:25:00



0:24:15 The opposite of irritability is escape mechanism. So this is where the higher pacemaker slows down or fails. So the S-A node gets sick, and it slows down at a rate lower than the A-V node, or it stops working completely, the A-V node is going to kick in at its inherent rate and now become the pacemaker. So an escape mechanism is where the higher pacemaker fails and the lower pacemaker has to kick in in order to maintain some level of cardiac output and blood pressure. The last slide in this lecture just shows us the impact of the sympathetic nervous system versus the parasympathetic nervous system.

14 Autonomic Nervous System

0:25:00 – 0:26:54



0:25:13 People that might have an increased parasympathetic input is somebody who is exposed to fertilizer, organic phosphates, which increase parasympathetic activity. So when parasympathetic activity is increased on the heart's electrical system, it's going to slow the heart rate down. It's going to slow down the force of contractility.

0:25:47 Not super important, not really going to test a lot on that. If you know somebody that doesn't do good with needle sticks or blood draws, sometimes they can have a vasovagal reaction, and that's increased parasympathetic input into the heart. And so those people might feel a little bit faint and their heart rate slows down, or they might actually faint. That would be another example of increased parasympathetic activity to the heart. Sympathetic activity to the heart will, so if you get scared, or you get nervous, or you get short of breath, or you're in distress, you're going to have increased sympathetic activity. This is known as fight or flight.

0:26:29 And in this situation, your heart rate's going to go faster. It's going to contract harder. But not super important. The previous slides are definitely important. This one, not as much. And that's going to conclude our second lecture for the EKG course.